

BUS23-2

Business Models & Value Creation

Decoding how value is created, captured, defended — and where it breaks

How does this business make money, why does that hold, and where is it structurally fragile?

Albert School — 22 hours

Preface

You arrive here with two reflexes already trained. From **Marketing Fundamentals** (BUS13-1) you learned that strategy is the discipline of choosing what **not** to do — which segments to ignore, which position to commit to, how to allocate a fixed budget across competing levers. From **Designing Operations from Scratch** (BUS13-2) you learned to ask of any plan: **can this actually be delivered, and where will it break first?** You have, in other words, met the demand side and the supply side of a business in the flesh.

This course gives you the connective tissue between them: the **business model** — the economic architecture that explains **how** a firm creates value for someone, **captures** some of that value as revenue and margin, and either **defends** that capture over time or quietly loses it. A business model is not a business plan, not a pitch, not a mission statement. It is a precise, falsifiable claim about who pays you, for what, why they keep paying, and what it costs you to serve them. When that claim is coherent, the business compounds. When one cell of it rests on an unexamined assumption, the whole structure is one bad quarter from collapse — and the job of this course is to teach you to find that cell **before** the market does.

We build the vocabulary in a deliberate order. First the **Business Model Canvas** as a questioning scaffold — not a form to fill, but a set of nine interrogations whose real value is exposing hidden assumptions and critical dependencies. Then the **typologies** (SaaS, marketplace, freemium, razor-and-blade, and the rest) and the viability condition each one lives or dies by. Then the **unit economics** — CAC, LTV, gross margin, contribution margin, break-even — defined here **once, precisely**, because every later course in the arc (Financial Diagnosis, Corporate Finance, Valuation, the whole Strategy sequence) will **mobilise** these numbers rather than re-deriving them. Then the deeper questions: why **value created** and **value captured** are different quantities and why some industries create fortunes they cannot keep; what actually makes a model **defensible** and when each supposed moat is an illusion; how to **stress-test** a model by doubling its CAC or compressing its margin and reading off the priority levers; and finally when a **pivot** is a genuine re-architecture rather than a rebrand of the same failure.

The course ends where it must: you take a real local business, deconstruct it end to end — canvas, typology, unit economics, stress test — and defend that deconstruction before a jury that may include the founder. That is the test of whether the vocabulary has become judgement. By the end you should be unable to hear “we’re a marketplace for X” without immediately asking which side is hard to get, what the take rate is, what a cohort’s LTV/CAC looks like, and what happens when acquisition cost doubles. That reflex is the whole point.

Contents

Preface	2
1 The business model canvas as a questioning scaffold	3
1.1 A juice bar that looked fine on paper	3
1.2 Nine blocks, read as questions	3
1.3 Read in logical order, not left to right	4
1.4 Assumptions, dependencies, and fragilities	5
2 Business model typologies	6
2.1 The same nine blocks, arranged for a different fight	6
2.2 A working taxonomy	6
2.3 Each typology has one viability condition	6
2.4 B2B2C: selling through a business to its customers	7
3 Fundamental unit economics	8

3.1	The number that settles the argument	8
3.2	Margins: what a sale actually contributes	8
3.3	CAC and LTV: the two-sided test of a customer	8
3.4	Break-even: the volume that pays the fixed costs	9
4	Value creation versus value capture	10
4.1	The industry that created everything and kept nothing	10
4.2	The value stick	10
4.3	Why creation and capture diverge	11
5	Defensibility: moats and their illusions	12
5.1	Why the second mover sometimes wins	12
5.2	The four classic mechanisms	12
5.3	When each moat is illusory	13
6	Stress-testing a business model	14
6.1	The model that worked only in fair weather	14
6.2	The two canonical shocks	14
6.3	Reading off the priority levers	14
7	Business model pivots	15
7.1	When changing the model is courage, and when it is denial	15
7.2	What a genuine pivot changes	16
8	Integrated deconstruction: putting it together	17
8.1	The skill the whole course was building toward	17
8.2	A protocol for deconstruction	17
8.3	Critiquing a canvas: assumed versus validated	17

1 The business model canvas as a questioning scaffold

1.1 A juice bar that looked fine on paper

Two friends open a cold-pressed juice bar. The pitch is crisp: premium juice, a prime high-street location, health-conscious office workers as customers, revenue from walk-in sales. Every box a plausible investor would want is ticked. Eighteen months later they are insolvent — and the reason was visible from day one to anyone who read the model as a **set of dependencies** rather than a **list of features**.

The prime location (a strength on the pitch) carried €9,000/month in rent. To cover it at a €6 juice with €4 of contribution, they needed roughly 1,500 juices a month **before paying themselves** — about 70 a day, every day, in a shop with a lunchtime rush and dead afternoons. The “health-conscious office workers” turned out to buy a juice twice a month, not twice a week. The model never failed at execution; it failed at **architecture**. The revenue side and the cost side rested on assumptions that, written next to each other, could not both be true. The Business Model Canvas is the tool that puts them next to each other.

1.2 Nine blocks, read as questions

The canvas, due to Osterwalder and Pigneur, decomposes any business model into nine interlocking blocks. Its power is not as a template to complete but as a **scaffold for interrogation**: each block is a question, and the interesting answers are the ones that contradict each other.

Definition 1 (the nine blocks)

- **Customer Segments** — for whom are we creating value, and who is the most important customer? Distinct segments have distinct needs, channels, and willingness to pay.
- **Value Propositions** — what bundle of products and services creates value for each segment, and which customer problem does it solve or job does it get done?
- **Channels** — through which touchpoints do we reach and deliver to each segment (awareness, purchase, delivery, after-sales)?
- **Customer Relationships** — what kind of relationship does each segment expect (self-service, personal, community, automated), and what does it cost to maintain?
- **Revenue Streams** — for what value is each segment **actually willing to pay**, how, and on what model (one-off, subscription, usage, licensing)?
- **Key Resources** — what assets are indispensable to make the model work (physical, intellectual, human, financial)?
- **Key Activities** — what must the firm do exceptionally well (produce, solve problems, run a platform)?
- **Key Partners** — who do we rely on outside the firm (suppliers, allies), and which activities or resources do they provide?
- **Cost Structure** — what does the model cost to run, and is it driven mostly by fixed or variable costs?

Key Partners	Key Activities	Value Propositions	Customer Rel.	Customer Segments
	Key Resources		Channels	
Cost Structure		Revenue Streams		

← how the model spends

how the model earns →

Figure 1: The Business Model Canvas. The right half (segments, value proposition, channels, relationships, revenue) is the **value and earning** side; the left half (resources, activities, partners, cost) is the **efficiency and spending** side. The two halves must balance: every promise on the right is paid for by something on the left.

1.3 Read in logical order, not left to right

Beginners fill the canvas top-left to bottom-right and end up with a tidy grid that hides the very contradictions it should expose. The disciplined reading follows the **logic of the business**, which always starts on the right:

A logical order for interrogating the canvas

1. **Customer Segments** — who, precisely, and which one matters most? Resist “everyone.”
2. **Value Proposition** — what job does it do for **that** segment, and why us over the alternative (including doing nothing)?
3. **Channels and Relationships** — how do we reach, win, and keep them, and at what cost?
4. **Revenue Streams** — what will they actually pay, and on what model?
5. **Key Resources, Activities, Partners** — what must be true on the supply side to deliver that proposition repeatedly?

6. **Cost Structure** — add it all up. Does the revenue side, at realistic volume, exceed the cost side?

Reading in this order forces the two halves to confront each other. The juice bar's canvas, read this way, breaks at step 6: the revenue the segment will realistically generate cannot cover the cost structure the location and format impose. That is a **structural** failure, not an execution problem — no amount of better marketing fixes a model whose two halves contradict.

1.4 Assumptions, dependencies, and fragilities

The canvas earns its place in this course because it makes three things explicit that a business plan hides.

Definition 2 (assumption, dependency, fragility) A **hidden assumption** is a claim a cell relies on that has not been validated — “office workers will buy juice twice a week.” A **critical dependency** is a cell whose failure takes the whole model down — a single supplier, one acquisition channel, a key partner. A **structural fragility** is a place where the architecture itself is brittle: two cells that can only both be true under optimistic conditions, or a revenue stream that rests entirely on one unvalidated assumption.

Common pitfall The most common canvas error is treating a filled cell as a **fact**. A cell you wrote from hope (“customers will pay €6”) and a cell you wrote from evidence (“our 200 paying customers pay €6”) look identical on the grid and are worlds apart in risk. We make this distinction the entire subject of Chapter 7 (assumed vs. validated). For now: every cell is a claim, and a claim is only as strong as the evidence behind it.

Worked example — Finding the critical dependency. A meal-kit startup's canvas looked diversified — three segments, two channels, a rich value proposition. But reading the **Key Partners** cell revealed that 80% of orders were acquired through a single paid-ads channel on one platform, and **Key Resources** depended on one regional fulfillment partner. Two cells, two single points of failure. When the ad platform changed its pricing, CAC doubled overnight (the exact shock of Chapter 6) and the model — diversified on paper — proved to hang from two threads. The canvas did not predict the event; it located the **fragility** that made the event fatal.

Exercises

1. Fill the nine blocks for a local café you know. Then re-read them in the logical order above and name the first place the right half and left half fail to balance.
2. For that café, mark each cell as **assumed** or **validated** given what you actually know. Which single unvalidated cell, if false, breaks the model?
3. Identify one critical dependency (a cell that is a single point of failure) and propose a concrete change to the model that reduces it.

2 Business model typologies

2.1 The same nine blocks, arranged for a different fight

Two software companies sell the same capability — invoicing tools for freelancers. One charges €15/month per user (SaaS subscription); the other gives the tool away free and takes 1% of every invoice paid through it (a transaction model). Identical software, **radically** different businesses: different CAC tolerances, different churn sensitivities, different definitions of “a good customer,” different reasons they might die. A **typology** is a recurring arrangement of the canvas that comes with its own physics — its own viability condition you must satisfy or perish.

2.2 A working taxonomy

Definition 3 (core business model typologies)

- **SaaS / subscription** — recurring fee for ongoing access to software or a service. Revenue is predictable and compounding; the enemy is **churn**.
- **Marketplace** — the firm intermediates between independent buyers and sellers, earning a **take rate** on transactions. Creates little itself; orchestrates others.
- **Freemium** — a free tier acquires users at near-zero marginal cost; a paid tier converts a fraction of them. Lives on the **conversion rate** from free to paid.
- **Razor-and-blade** — sell the durable good cheap (or at a loss), profit on the consumable repurchased forever (printers/ink, console/games). Lives on **lock-in** to the consumable.
- **Asset-light** — own the customer relationship and brand, not the productive assets (Uber owns no cars, Airbnb no rooms). Scales fast; defensibility is the question.
- **B2B / B2C / B2B2C** — not a revenue model but a **customer-architecture**: selling to businesses, to consumers, or **through** a business to its consumers.

2.3 Each typology has one viability condition

The reason to classify is not taxonomy for its own sake — it is that each type fails in a characteristic way, and naming the type tells you the **one number** to watch.

Theorem 4 (viability conditions by type)

- **SaaS / subscription** is viable only if customers stay long enough to repay acquisition: $\frac{LTV}{CAC}$ comfortably above 1 (rule of thumb ≥ 3), which demands **low churn**.
- **Marketplace** is viable only if it solves the **chicken-and-egg** cold-start (no buyers without sellers, no sellers without buyers) and its **take rate** is high enough to fund operations yet low enough that neither side disintermediates it.
- **Freemium** is viable only if (free-tier cost per user) $\times$ (users per paying customer) stays well below the paying customer’s contribution — i.e. the **conversion rate** covers the cost of the free riders.
- **Razor-and-blade** is viable only if the blade is **protected** (patent, lock-in, brand) — an unprotected blade is copied by a generic and the subsidised razor becomes a pure loss.
- **Asset-light** is viable only if the orchestration layer is **defensible**; owning no assets also means owning no barrier, so it must build one (network effects, brand, data).

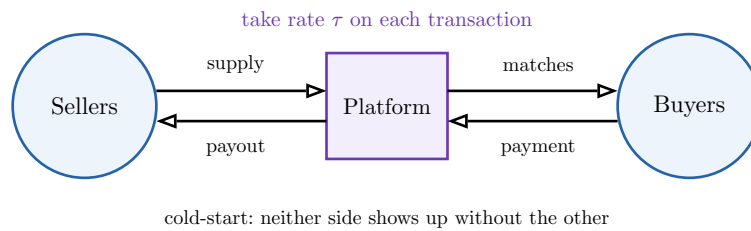


Figure 2: A marketplace intermediates two interdependent sides and earns a take rate τ on the value flowing across. Its two existential problems are visible: the cold-start (bootstrapping both sides at once) and disintermediation (either side going around the platform once matched).

Common pitfall Do not confuse the **revenue model** with the **customer architecture**. “We’re B2B” answers **who buys**; “we’re SaaS” answers **how we charge**. A firm can be B2B SaaS, B2C subscription, or B2B2C marketplace. Collapsing these two axes into one label (“we’re a SaaS”) hides half the model — and the half you hid is usually where the viability condition lives.

2.4 B2B2C: selling through a business to its customers

The B2B2C architecture deserves its own note because students routinely mis-model it. A payments provider that sells to shops, which in turn serve consumers, has **two** customers with **two** value propositions and only **one** of them pays directly. The viability question becomes: who has the power in the chain, and who owns the end-customer relationship? A business that reaches consumers only through a partner that could replace it is structurally fragile no matter how good its product.

Worked example — Remodelling one business under three types. Take a fitness studio. As **B2C subscription**: members pay €80/month, viability hinges on churn and class utilisation. As a **marketplace** (asset-light): the studio becomes a platform listing independent instructors’ classes, taking 20% — now the fight is supply (good instructors) and the cold-start. As **B2B2C**: it sells corporate wellness packages to employers who offer classes to staff — one big buyer, very different CAC and churn, but dangerous dependency on a few corporate contracts. Same room, same instructors, **three** different businesses — different unit economics, different fragilities, different reasons to fail. This is exactly the remodelling exercise the OP demands (LO 3).

Exercises

4. Classify five businesses you use this week by typology, and for each name the single viability number its survival depends on.
5. Take one local business and remodel it under three different typologies. For each, state what changes in the activity system, the unit economics, and the customer relationship.
6. A razor-and-blade hardware startup sells its device at a €40 loss, expecting €5 profit per monthly consumable. A generic consumable appears. Explain precisely why the whole model — not just the consumable line — is now underwater.

3 Fundamental unit economics

3.1 The number that settles the argument

A founder insists her subscription box is “growing fast — 40% month over month.” Growth hides everything. The question that settles whether the growth is **worth having** is one ratio: does a customer, over their whole life with you, generate more contribution than it cost to acquire them? If yes, growth compounds into a fortune; if no, every new customer **accelerates** the bankruptcy. Unit economics is the arithmetic of that single question, and this course defines it **once, precisely**, because every downstream course will use these exact definitions without re-deriving them.

3.2 Margins: what a sale actually contributes

Definition 5 (gross margin and contribution margin) **Gross margin** is the fraction of revenue left after the **cost of goods sold** (COGS — the direct cost of producing what you sold):

$$\text{gross margin} = \frac{\text{revenue} - \text{COGS}}{\text{revenue}}.$$

Contribution margin per unit is price minus **all variable cost** of one unit:

$$\text{CM} = \text{price} - \text{variable cost per unit},$$

the cash each sale contributes toward fixed costs and profit. Gross margin is a **percentage** describing the model’s quality; contribution margin is a **per-unit amount** you spend against (Chapter 8 of operations called it the per-unit budget).

A high gross margin (software: 80–90%) means almost every euro of new revenue is available to fund growth or fall to profit; a low one (grocery retail: 20–25%) means the model must win on volume and operational discipline. Gross margin, more than any single number, sets what kind of business you are allowed to build on top.

3.3 CAC and LTV: the two-sided test of a customer

Definition 6 (CAC) **Customer Acquisition Cost** is the fully-loaded cost of acquiring one new customer over a period:

$$\text{CAC} = \frac{\text{total sales \& marketing spend}}{\text{number of new customers acquired}}.$$

“Fully-loaded” means **all** of it — ad spend, salaries of the sales and marketing team, tools, commissions — not just the media bill. Understating CAC by counting only ad spend is the most common way a doomed model looks healthy.

Definition 7 (LTV) **Lifetime Value** (also LTV or CLV) is the total **contribution** a customer generates over their entire relationship with the firm. For a subscription with roughly constant spend:

$$\text{LTV} = \frac{\text{ARPU} \times \text{gross margin}}{\text{churn rate}},$$

where **ARPU** is average revenue per user per period and **churn rate** is the fraction of customers lost per period. The $\frac{1}{\text{churn}}$ factor is the average customer **lifetime** in periods: at 5% monthly churn the average customer stays $\frac{1}{0.05} = 20$ months.

The two ideas fuse into the ratio that the whole subscription economy lives by:

Theorem 8 (the LTV/CAC test) A model is viable only if a customer is worth more than they cost:

$$\frac{\text{LTV}}{\text{CAC}} > 1.$$

The working rule of thumb is $\frac{\text{LTV}}{\text{CAC}} \geq 3$ — below 1 you lose money on every customer and growth is fatal; near 1 there is no margin for error; **far above 3** often signals you are **under-investing** in growth and leaving the market to a competitor. A second, complementary number is the **CAC payback period** — months of contribution to recover CAC — which matters because a great lifetime ratio is useless if you run out of cash waiting for it.

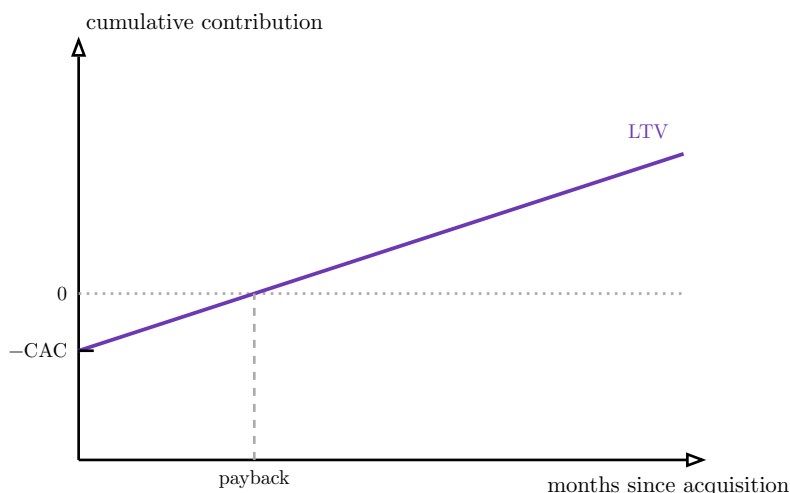


Figure 3: The cohort payback curve. A customer starts the relationship “in the hole” by $-\text{CAC}$; each period of contribution climbs the line. It crosses zero at the **payback period** and keeps rising to the total **LTV**. A healthy model crosses zero fast and ends far above it; a doomed one never crosses, or crosses so late the firm runs out of cash.

3.4 Break-even: the volume that pays the fixed costs

Definition 9 (break-even volume) **Break-even volume** is the number of units at which total contribution exactly covers fixed costs:

$$\text{break-even volume} = \frac{\text{fixed costs}}{\text{contribution margin per unit}}.$$

Below it the firm loses money; above it, each additional unit’s full contribution falls to profit. This is the bridge from unit economics (per-customer) to the whole firm’s P&L.

Common pitfall The textbook errors in unit economics are arithmetic-but-fatal. **Counting revenue, not contribution, in LTV** inflates a customer's worth by the whole COGS — a 20%-margin business that uses revenue overstates LTV fivefold. **Understating CAC** by excluding salaries flatters every model. **Using blended CAC** (new customers $\div$ total spend including customers who'd have come for free) hides that **paid** acquisition is unprofitable. And **ignoring payback** lets a business with great lifetime economics die of thirst long before the lifetime arrives. Define each term the same way every time — that consistency is precisely why this course pins the definitions down once.

Worked example — Two businesses, one looks better, the other is. Company A: ARPU €50/month, 70% gross margin, 8% monthly churn, CAC €600. Company B: ARPU €50/month, 70% gross margin, 3% monthly churn, CAC €700. A's $LTV = \frac{50 \times 0.7}{0.08} = €437$ — **below its €600 CAC**: it loses €163 per customer and growth is suicide. B's $LTV = \frac{50 \times 0.7}{0.03} = €1,167$ against €700 CAC: ratio 1.67, payback $\frac{700}{50 \times 0.7} \approx 20$ months. B is still not great (ratio below 3, slow payback) but it is **alive**; A is dead and growing. The only structural difference is churn — which is why the SaaS viability condition is, at root, a churn condition.

Exercises

7. A SaaS product has ARPU €30/month, gross margin 80%, monthly churn 4%, and CAC €450. Compute LTV, LTV/CAC, and the CAC payback period. Is it viable? What is the single most effective lever?
8. A hardware unit sells for €120 with €78 variable cost; fixed costs are €60k/month. Find the contribution margin, gross margin, and monthly break-even volume.
9. Show numerically how counting **revenue** instead of **contribution** in LTV changes the verdict for a business with a 25% gross margin.
10. A founder reports CAC of €40 using only ad spend. The sales/marketing team costs €20k/month and acquires 300 customers/month from all sources. What is the fully-loaded CAC, and why does the gap matter?

4 Value creation versus value capture

4.1 The industry that created everything and kept nothing

Commercial aviation has, over its history, created staggering value — it moved billions of people, knit economies together, made the modern world possible. It has also, in aggregate, been a **terrible** business: cumulative industry profits are famously close to zero or negative. Meanwhile a humble maker of in-flight pretzels or a single congested-airport slot can be wildly profitable. The lesson is the course's most important conceptual distinction: **the value a business creates and the value it captures are two different quantities, and they can diverge enormously.**

4.2 The value stick

The cleanest way to see the gap is the **value stick** — a vertical bar from what a customer would maximally pay down to what a supplier would minimally accept.

Definition 10 (the value stick)

- **Willingness to pay** (WTP) — the most a customer would pay before walking away.
- **Price** — what the customer actually pays.

- **Cost** — what the firm pays its suppliers for inputs and labour.
- **Willingness to sell (WTS)** — the least a supplier (or employee) would accept.

Value created = $WTP - WTS$: the total surplus the transaction brings into the world. It is split three ways:

$$\underbrace{WTP - \text{price}}_{\text{customer's share}} \quad \underbrace{\text{price} - \text{cost}}_{\text{firm captures}} \quad \underbrace{\text{cost} - WTS}_{\text{supplier's share}}$$

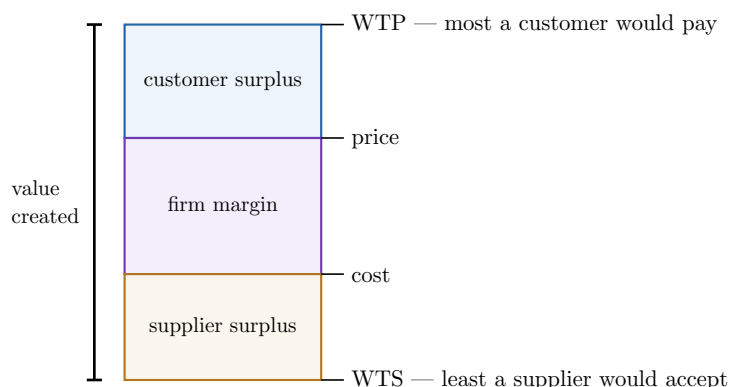


Figure 4: The value stick. Total value created ($WTP - WTS$) is split among customer, firm, and supplier. A firm can create huge value (a tall stick) yet capture little (a thin middle band) if competition pushes price toward cost or powerful suppliers push cost toward WTP. Strategy is the art of **widening the middle band**.

4.3 Why creation and capture diverge

Capture is governed not by how much value you create but by **bargaining power** and **competition**. A firm captures a wide middle band only when something stops price from falling to cost and stops input cost from rising to WTP.

Theorem 11 (the determinants of capture) A firm captures value to the degree that: (i) **competition is limited**, so rivals cannot bid price down toward cost; (ii) **customers lack alternatives or power**, so price can sit well below their WTP; (iii) **suppliers and employees lack power**, so cost sits well above their WTS. Where all three pressures are intense — a commodity sold by many firms to informed buyers using scarce inputs — value created can be enormous and value captured near zero. The defensibility mechanisms of the next chapter are precisely the structures that hold these three pressures off.

Common pitfall Founders systematically conflate “we create a lot of value” with “we will make a lot of money.” They are different claims. Open-source software, free consumer apps, and much of journalism create immense value and capture little. Before celebrating how much good your product does, ask the colder question: **what stops the price you can charge from collapsing to your cost?** If you have no answer, you are running an aviation business.

Worked example — Same value, opposite capture. Two firms make GPS chips of identical quality. One sells into a market of a dozen interchangeable suppliers bidding for smartphone makers’ contracts — price is hammered to just above cost, capture is razor-thin

despite creating real value in every phone. The other holds a patent on a navigation algorithm no one can legally replicate — same value per chip, but competition cannot bid the price down, so the middle band is wide. The difference is not the value created (identical) but the **barrier** protecting capture. Value creation buys you the right to play; value capture is what you actually take home.

Exercises

11. Draw the value stick for a product you buy regularly. Estimate where price and cost sit, and say who captures the most surplus and why.
12. Name an industry that clearly creates enormous value but captures little. Identify which of the three pressures (competition, customer power, supplier power) is doing the damage.
13. For a named business, describe one concrete move that would **widen its captured band** without changing the value it creates.

5 Defensibility: moats and their illusions

5.1 Why the second mover sometimes wins

A startup invents a beloved product, grows fast — and is then crushed by a copycat with more capital. Another startup, no more clever, becomes unassailable. The difference is **defensibility**: whether the model contains a structural reason that capture, once won, is hard to take away. Value capture (Chapter 4) is a snapshot; defensibility is whether the snapshot **holds**. Crucially, most things founders call moats are not, and learning to tell a real moat from an illusory one is the heart of this chapter.

5.2 The four classic mechanisms

Definition 12 (defensibility mechanisms)

- **Network effects** — the product becomes **more valuable to each user as more users join** (a telephone, a marketplace, a social network). The barrier is that a rival must replicate the whole network, not just the product.
- **Switching costs** — it is costly (money, effort, risk, data) for a customer to leave, so they stay even when a rival is marginally better (an ERP system, a bank account).
- **Scale economies** — unit cost falls with volume, so the largest player profitably underprices entrants (manufacturing, logistics, cloud infrastructure).
- **Proprietary data** — the firm accumulates data from use that improves the product in a way rivals cannot match, creating a **data feedback loop**.

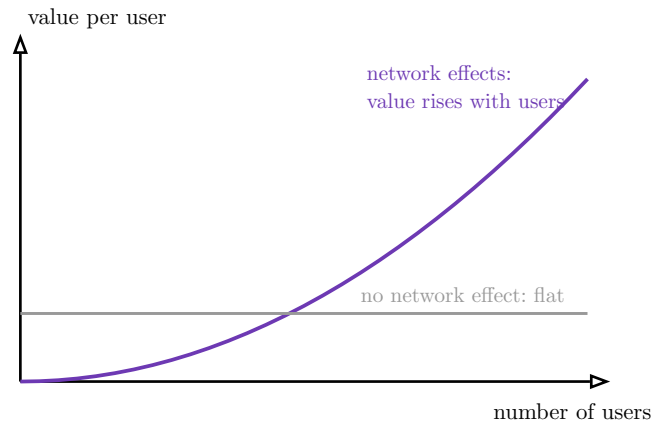


Figure 5: A genuine network effect: each additional user raises the value every other user gets, so the curve bends upward and the leader pulls away. A product whose value is flat in the number of users (a better mousetrap, however good) has **no** network effect — merely growth, which a funded rival can buy.

5.3 When each moat is illusory

The danger is that each mechanism has a convincing counterfeit. Naming the counterfeit is the skill the OP asks for.

Theorem 13 (when the moat is illusory)

- **Network effects are illusory** when they are **local or weak** (value plateaus after a few users), **multi-homing** is easy (users belong to several platforms at once — food-delivery couriers and diners use them all), or growth is mistaken for an effect (more users but no extra value **per** user).
- **Switching costs are illusory** when migration tools, importers, or regulation (data portability) cut them, or when the pain is one-time and small relative to a rival’s ongoing advantage.
- **Scale economies are illusory** when minimum efficient scale is **low** (a small player is already cost-competitive) or when costs are mostly variable (no leverage from volume).
- **Proprietary data is illusory** when the data does not actually **compound the product’s value**, when the same data is cheaply available elsewhere, or when commodity models reach “good enough” without it.

Common pitfall “We have a network effect” is the most over-claimed sentence in startup pitches. Test it with one question: **does an additional user make the product better for existing users, or just add one more user?** A gym is not a network — your membership’s value does not rise because a stranger joined; it may **fall** (crowding). Many “network effect” businesses are simply businesses with customers. Real network effects are rare, which is exactly why they are valuable.

Worked example — Two food-delivery moats, one real. A food-delivery app claims two moats. First, “network effects: more diners attract more restaurants.” Partly real but **weak** — diners and restaurants **multi-home** across three apps, so no single app pulls decisively ahead, which is why the category stays a bloody, unprofitable war. Second, “proprietary data on delivery times improves routing.” This one **compounds**: better routing → faster delivery → more orders → more routing data → still better routing, a loop a small rival cannot match. The

defensible moat was the unglamorous data loop, not the much-touted network effect. Telling them apart is the whole game.

Exercises

14. For a platform you use, state which of the four moats it claims, then apply the illusory-test to each. Which survive?
15. Explain why “we grew 50% last quarter” is evidence of demand but **not** of a network effect. What additional evidence would demonstrate a real one?
16. Give a real example of a switching cost that **was** a strong moat until a migration tool or regulation dissolved it.

6 Stress-testing a business model

6.1 The model that worked only in fair weather

A model that is profitable today can be one shock away from ruin, and the shocks come from two predictable directions: **acquisition gets more expensive** or **margin gets squeezed**. A disciplined deconstruction does not stop at “the unit economics work **now**”; it asks “what happens when they don’t?” — and then reads off which levers to pull. This is the operational **failure-mode analysis** from BUS13-2 applied to the economic architecture instead of the physical one.

6.2 The two canonical shocks

Definition 14 (the standard stress tests)

- **CAC doubling** — acquisition cost rises $\times 2$ (an ad platform re-prices, a channel saturates, a competitor bids up the auction). Test: does $\frac{LTV}{CAC}$ stay above 1? Does payback stay inside the cash runway?
- **Margin compression** — gross margin falls by 10 percentage points (input costs rise, price competition, a needed discount). Test: does contribution still cover CAC and fixed costs, and how far does break-even volume move?

These two are canonical because they are the **most common** ways real models die and because they probe the two halves of the canvas — the earning side (CAC, the revenue engine) and the spending side (margin, the cost structure).

Worked example — Running the CAC-doubling shock. Recall Company B from Chapter 3: LTV €1,167, CAC €700, ratio 1.67, payback 20 months. Double the CAC to €1,400. Now $LTV/CAC = \frac{1167}{1400} = 0.83$ — **below 1**. The business that was merely mediocre is now **destroying value on every new customer**, and payback exceeds the average customer’s lifetime, so it never recovers the spend. A model with a ratio of 5 would have dropped to 2.5 and survived comfortably. This is **why** the rule of thumb wants headroom far above 1: the cushion is not greed, it is shock-absorption.

6.3 Reading off the priority levers

The point of the stress test is not the verdict but the **levers** — and the model itself tells you which three matter most.

Theorem 15 (the priority-lever logic) Under either shock, the levers that restore viability are, in rough order of controllability: (1) **reduce churn** — it multiplies LTV directly via the $\frac{1}{\text{churn}}$ factor and is often the highest-leverage move; (2) **raise ARPU or price** — straight into the numerator of LTV and contribution; (3) **improve gross margin** — lower COGS or shift mix; (4) **lower CAC** — better targeting, cheaper channels, more organic acquisition. The **three priority levers** for a given business are the subset that (a) move the binding number most and (b) are actually within reach given the model’s constraints. Naming them **with the arithmetic**, not by intuition, is the deliverable.

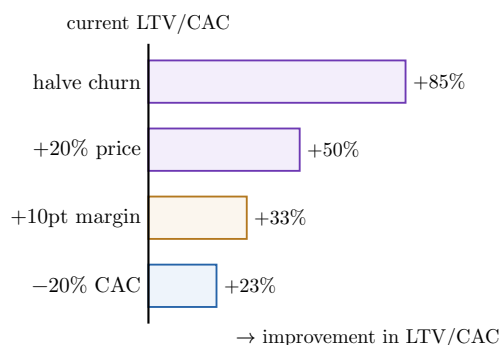


Figure 6: A lever chart for one business under stress. Halving churn moves the viability ratio most (it compounds through the customer lifetime), then price, then margin, then CAC. The **priority levers** are read off the longest bars that the business can actually pull — here, churn and price.

Common pitfall Two stress-test errors recur. First, testing one shock in isolation when reality delivers **both at once** — a downturn raises CAC **and** compresses margin together; a model must survive the joint shock, not just each alone. Second, choosing levers by **ease** rather than **impact** — “we’ll cut CAC” feels actionable but if churn is the binding constraint, trimming CAC rearranges deckchairs. Let the arithmetic, not the comfort, pick the levers.

Exercises

17. Take the SaaS product from the Chapter 3 exercises (ARPU €30, margin 80%, churn 4%, CAC €450). Apply a CAC doubling. Compute the new LTV/CAC and state whether it survives.
18. Apply instead a 10-point margin compression (80% → 70%) to the same product. Which shock hurts more, and why?
19. For that product, rank the four levers by their effect on LTV/CAC and name the three you would prioritise, with the arithmetic justifying each.
20. Model both shocks simultaneously. What is the minimum churn reduction that would keep the joint-shocked model above LTV/CAC = 1?

7 Business model pivots

7.1 When changing the model is courage, and when it is denial

A startup is not working. The founders “pivot.” Sometimes that pivot saves the company by re-architecting how value is created and captured; sometimes it is the same failing model wearing

new clothes, and it merely delays the funeral. The discriminating question is **structural**: did the economic architecture actually change, or only the label?

7.2 What a genuine pivot changes

Definition 16 (pivot) A **pivot** is a structured change to one or more core elements of the business model while preserving the accumulated learning. A **genuine** pivot changes the **architecture** — a different customer segment, value proposition, revenue model, or key activity — such that the **unit economics or the viability condition themselves change**. A **cosmetic** pivot (a “rebrand of the same failure”) changes messaging, naming, or surface features while the underlying canvas — and therefore the reason it was failing — stays intact.

Common genuine pivots (each changes a different canvas cell):

- **Customer-segment pivot** — same product, a different buyer for whom the value is real.
- **Value-proposition pivot** — a feature customers actually loved becomes the whole product.
- **Revenue-model pivot** — same value, a different way of charging (one-off → subscription, paid → freemium) that fixes the unit economics.
- **Platform pivot** — from a single application to a platform others build on, or vice versa, changing the activity system and defensibility.

Theorem 17 (the pivot test) A change is a genuine pivot if and only if it alters the model’s **binding viability condition** — the churn that was too high, the CAC that exceeded LTV, the cold-start that never resolved. If after the change the **same number** still binds and is still failing, the pivot was cosmetic. The honest test: write the unit economics before and after. If the ratio that was broken is broken in the same way, you rebranded; if a **different, healthier** arithmetic now governs the business, you pivoted.

Common pitfall The seductive failure is the **serial cosmetic pivot**: a team that re-skins, re-names, and re-pitches every few months, each time feeling like decisive action, never once changing the cell of the canvas that is actually broken. Activity is mistaken for re-architecture. Before pivoting, name the **specific** cell and the **specific** number that must change; if the proposed move does not touch it, it is motion, not progress.

Worked example — A real re-architecture vs. a rebrand. A company built a video-game-streaming product that no one paid for — CAC far exceeded LTV because consumers churned in weeks. **Cosmetic attempt**: new name, new logo, influencer marketing. CAC and churn unchanged; still dead. **Genuine pivot**: they noticed their **broadcasting tool**, built as a side feature, was what users actually returned for, and the real buyers were professional streamers, not casual gamers. They cut the game product entirely and sold the tool on subscription to creators. New segment, new value proposition, new revenue model — churn collapsed and LTV/CAC inverted. Same team, same code-base; a **different binding condition** now governed the business. That is the line between a pivot and a paint job.

Exercises

21. Find a well-known company that pivoted. State which canvas cell(s) changed and which viability number the pivot fixed.
22. Give an example (real or constructed) of a cosmetic pivot and show, via the before/after unit economics, why the same number was still binding.
23. A subscription business with 12% monthly churn proposes a pivot to a one-off purchase model. Argue whether this is genuine, and what new viability condition would then bind.

8 Integrated deconstruction: putting it together

8.1 The skill the whole course was building toward

Every chapter so far has been one instrument. The final deliverable plays them together: take a **real** company — ideally a local one whose founder you can question — and produce a complete deconstruction that maps the canvas, classifies the typology, computes the unit economics, stress-tests the model, and judges its defensibility, **then defend it before a jury that may include the founder**. The jury matters: a deconstruction that cannot survive the founder’s “no, that’s not how it works” was built on assumptions you mistook for facts.

8.2 A protocol for deconstruction

The deconstruction protocol

1. **Canvas.** Fill the nine blocks from evidence, marking each cell **assumed** or **validated**. State the hidden assumptions, the critical dependencies, and the structural fragilities (Chapter 1).
2. **Typology.** Classify the model and name the **one viability condition** that type lives or dies by (Chapter 2).
3. **Unit economics.** Compute CAC, LTV, LTV/CAC, gross and contribution margin, and break-even on **real data**, and compare each against the viability thresholds for the type (Chapter 3).
4. **Value creation vs. capture.** Identify what value the firm creates and the mechanism by which it does — or fails to — capture it (Chapter 4).
5. **Defensibility.** Name the claimed moats and apply the illusory-test to each (Chapter 5).
6. **Stress test.** Run CAC-doubling and margin-compression, jointly, and read off the three priority levers (Chapter 6).
7. **Verdict.** State where the model is structurally fragile and what evidence would move its riskiest assumed cell to validated (Chapter 1’s assumed-vs-validated, made actionable).

8.3 Critiquing a canvas: assumed versus validated

The OP singles out one analytic move (LO 7) as worth isolating: auditing which cells of a canvas are **assumed** and which are **validated**, and naming the evidence that would promote each.

Definition 18 (assumed vs. validated cell) A canvas cell is **validated** when there is real evidence it holds — paying customers, measured churn, observed CAC, signed partners. It is **assumed** when it rests on belief, analogy, or hope. **Critiquing a canvas** means classifying every cell on this axis and, for each assumed cell, naming the **specific, cheap experiment** whose result would move it to validated.

Worked example — Promoting a cell from assumed to validated. An early venture’s canvas asserted, in **Revenue Streams**, that “SMEs will pay €99/month.” That is an **assumed** cell — no SME had yet paid anything. The €99 was an anchor pulled from a competitor, not a measurement. The evidence that would validate it is cheap and specific: a pre-sales test offering the product at €99 to twenty qualified SMEs and counting how many actually pay, not how many say “interesting.” Until that test runs, the entire revenue side of the model — and every unit-economics number computed from it — is a **hypothesis dressed as a spreadsheet**. The jury’s first question is always: which of these numbers did you **measure**, and which did you **wish**?

Common pitfall The capstone error is **false precision**: computing LTV to the cent from an ARPU and a churn you assumed rather than measured. Numbers carry an authority their inputs have not earned. A deconstruction is honest only when each number is tagged with the quality of its source — measured, benchmarked, or guessed — and the verdict leans on the measured ones. Defending before a founder is the ultimate audit of exactly this honesty.

Exercises

24. Choose a real local business. Produce the full deconstruction following the seven-step protocol, tagging every unit-economics input as measured, benchmarked, or guessed.
25. For that business, identify the single riskiest **assumed** cell and design the cheapest experiment that would validate or refute it.
26. Anticipate the three hardest questions the founder would ask of your deconstruction, and prepare an evidence-based answer to each.
27. Write the one-paragraph verdict: the model’s typology, its binding viability number, its behaviour under the joint stress test, and the single structural fragility most likely to end it.

The architecture, in one paragraph. A business model is a falsifiable claim about how a firm creates value, captures some of it, and defends that capture. Map it with the canvas — nine blocks read as questions, every cell tagged assumed or validated — to surface hidden assumptions, critical dependencies, and structural fragilities. Classify its typology and watch the one number that type lives by. Compute the unit economics **once and precisely**: contribution and gross margin, CAC, $LTV = \frac{ARPU \times \text{margin}}{\text{churn}}$, the LTV/CAC test and payback, and break-even. Remember that value **created** and value **captured** diverge, governed by competition and bargaining power, and that defensibility — network effects, switching costs, scale, data — is real only when it survives the illusory-test. Stress-test with a doubled CAC and a compressed margin, jointly, and read off the three priority levers from the arithmetic. Distinguish a genuine pivot (the binding condition changes) from a rebrand (it does not). Then deconstruct a real business end to end and defend it before the people who know the truth. Three questions, always: **How does this make money, why does that hold, and where is it structurally fragile?**